EDGE-AI BASED PREDICTIVE CLOSED CONTROL SYSTEM FOR SMART POULTRY FARMING

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Presentation Outline

- Motivation
- Introduction
- Problem Statement & Objectives
- Literature Review
- Scope Of Project
- Project Applications
- Methodology
- Result
- Analysis
- Future Enhancements
- Conclusion
- References

Motivation

- Poultry farms require continuous environmental monitoring.
- Manual monitoring is labor-intensive and reactive.
- Uncontrolled environmental levels affect birds’ health and growth.
- Existing systems lack predictive intelligence.
- Edge-AI enables predictive, real-time environmental control.

Problem Statement & Objectives

Problem Statement:

- Open-house farms rely on manual observation.
- Existing IoT systems react only after thresholds break.
- Delayed response raises chick mortality risk.

Objective:

- To design and develop an IoT-based Edge-AI system for real-time monitoring, prediction, and control of environmental conditions in poultry houses, with a focus on the brooding stage of broiler production.

Literature Review

Paper	Author	Objective	Methodology	Findings	Limitations
IoT Closed House Control (2024) [4]	Surateno S. et al.	IoT-based environmental control for closed poultry houses	Threshold-based control with real-time sensors	Automated control of fans and heaters	Reactive only; no prediction capability
Broiler Temperature Needs (2013) [6]	Cassuce D. et al.	Determine optimal temperature requirements for broilers	Statistical analysis in controlled chambers	Week 1: 32-35℃; Week 4+: 20-22℃	No adaptive real-time control implementation
NH3 Impact on Growth (2004) [5]	Miles D. et al.	Analyze ammonia effects on broiler performance	Dose-response statistical modeling	6% weight loss, mortality 13.9% at 75 ppm NH3 (vs. 5.8% at 0 ppm)	Retrospective analysis only; no prevention mechanism
MQ-137 Validation (2025) [7]	da Rocha Balthazar G. et al.	Validate MQ-137 sensor for poultry ammonia monitoring	Sensor calibration in laboratory and poultry house	R=0.82, R^2=0.64 correlation	Monitoring-only; no ML integration
NH3 Risk Prediction (2024) [3]	Barbosa L. et al.	Predict ammonia concentration risk levels in poultry houses	MLP vs DT vs NB vs KNN on commercial farm data	MLP achieved >98% accuracy	No edge deployment; no real-time control

Gap Analysis

Limitations of Existing Solutions:

- **Reactive Control:** Surateno (2024) and similar IoT systems only respond *after* thresholds are breached, exposing chicks to harmful conditions.
- **No Edge-AI:** Barbosa (2024) achieved accurate MLP predictions but on desktop CPUs — no real-time, on-device inference.
- **No Closed-Loop Integration:** Balthazar (2025) validated MQ-137 but monitoring-only; no automatic actuation.
- **Age-Blind Control:** Most systems ignore broiler age, despite Cassuce (2013) proving temperature needs change weekly.

Research Gap Addressed by Our Project

Our system integrates:

- **TinyML (MLP)** for predictive ammonia and temperature spike detection.
- **ESP32-S3 edge deployment** for real-time inference (no cloud dependency).
- **Age-adaptive week-based thresholds** (Weeks 1-4+ per Cassuce 2013).
- **Closed-loop control** activating ventilation/heating before critical limits.
- **ESP-NOW communication** for reliable farm-area networking.

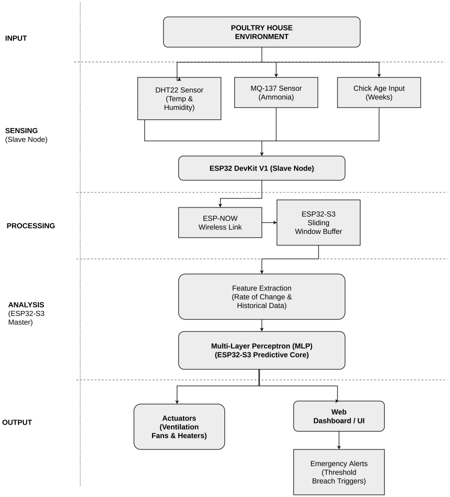
Scope Of Project

- Targets commercial broiler poultry farms.
- Real-time NH_3 and temperature data acquisition.
- Edge AI prediction using MLP on ESP32.
- Automated ventilation and alarm actuation on prediction.
- Web-based dashboard for remote farm monitoring.

Project Applications

- Continuous air quality monitoring in poultry farm houses.
- Early hazard detection to prevent bird mortality losses.
- Automated ventilation control for optimal farm environment.
- Scalable deployment across multiple poultry farm units.
- Data logging for farm productivity and environmental factors.
- Applicable to livestock and greenhouse monitoring systems.

Methodology -[1] System Block Diagram



Methodology - Input Layer (Data Acquisition)

- DHT22 sensor captures ambient temperature (℃) and relative humidity (%).
- MQ-137 sensor measures ammonia (NH_3) concentration in ppm.
- Chick age (in weeks) supplied so the model learns age-dependent patterns rather than fixed thresholds.
- Slave node (ESP32 DevKit V1) validates readings and forwards them continuously to the master.

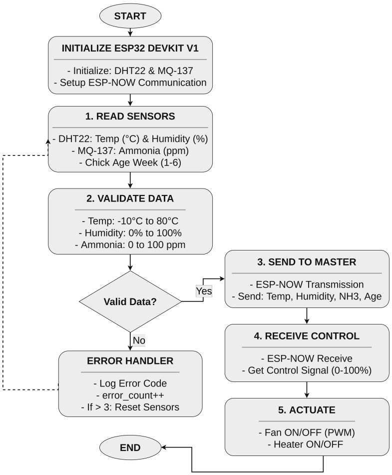
Methodology - Processing Layer (Prediction & Decision)

- Sliding window input fed into MLP.
- MLP forecasts future NH_3 and temperature values.
- The model runs on ESP32 via TFLite Micro inference engine.
- All processing happens on-device.

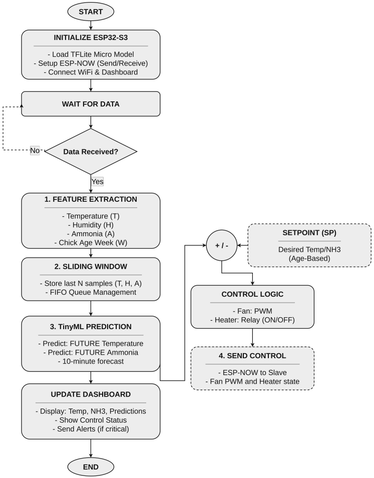
Methodology - Output Layer (Actuation & Monitoring)

- Ventilation activates through ML prediction or reactive hazard detection.
- Heating element triggered on low temperature prediction.
- Web dashboard displays live and predicted sensor values.
- The operator receives real-time alert messages on hazards.
- Events logged for future farm data analysis.

Methodology -[2] Slave Node Flowchart



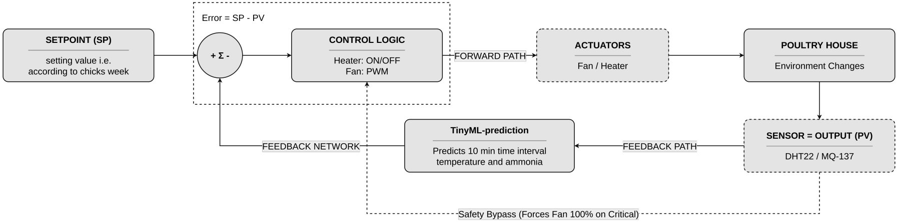
Methodology -[3] Master Node Flowchart



Methodology - [4] (Machine Learning Architecture)

- Bifurcated multi-task MLP — temperature and ammonia branches.
- 49 sliding-window features, shared 64-neuron Dense layer.
- Temperature branch: 32 neurons regression + classification.
- Ammonia branch: 32 neurons, extra 16-neuron classifier.
- Dropout and L2 Regularization used to reduce overfitting.
- TFLite quantized, 21.55 KB, runs on ESP32-S3.

Methodology -[5] Closed-Loop Control Flowchart



Methodology - [6] (Hardware Requirement)

- ESP32 DevKit V1 (Slave) & ESP32-S3 (Master)
- DHT22 (temp/humidity) & MQ-137 (ammonia) sensors
- Relay module for fan/heater switching
- LM2596 buck converter & 16$\times$2 I2C LCD
- DC cooling fan & PTC heating element
- 12V DC power supply

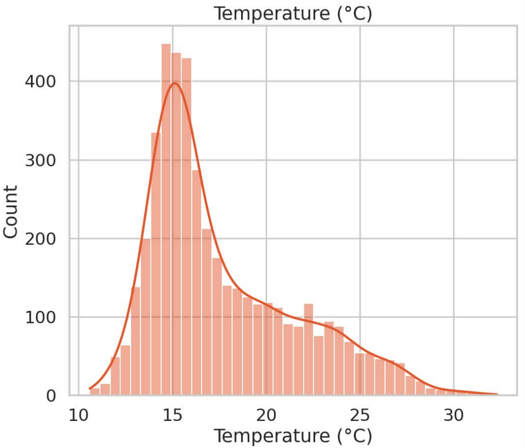
Methodology - [7] (Tools and Technologies)

- **Languages:** Python, C++, JavaScript
- **ML Stack:** TensorFlow, Keras, TFLite Micro, Scikit-learn
- **Backend:** Django, REST API, SQLite
- **Firmware:** PlatformIO, Arduino IDE
- **Dev Tools:** VS Code, Git, Jupyter Notebook
- **PCB Design:** KiCad

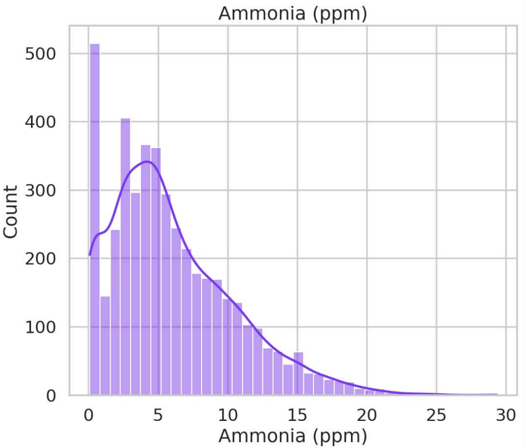
Methodology - [8] (Dataset Preparation)

- Source: naturally ventilated poultry-house field data (IEEE Dataport & PLOS ONE).
- 9,708 raw sensor readings collected over about 34 days.
- Cleaned and resampled to 4,458 uniform 10-minute records.
- Chronological 80/20 split: 3,041 train / 533 val / 884 test.
- 49 engineered features: lags, rolling stats, hour cycle, ammonia rate-of-change.
- Targets normalized with StandardScaler fit on training data only.

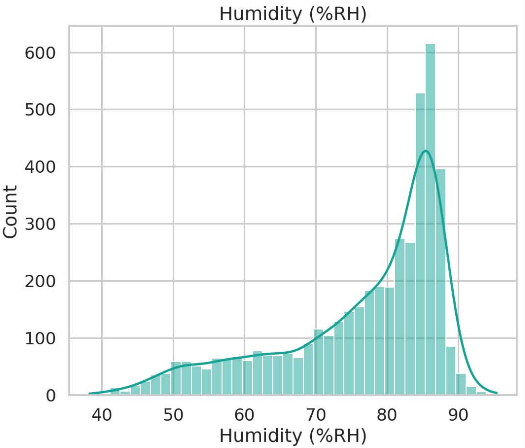
Methodology - [9] (Dataset Distribution Temperature)



Methodology - [10] (Dataset Distribution Ammonia)



Methodology - [11] (Dataset Distribution Humidity)



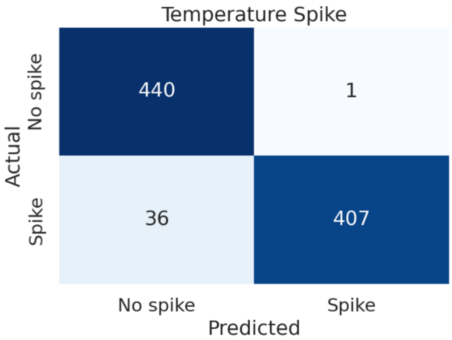
Methodology - [12] (Dataset Distribution Table)

Variable	Mean	Std Dev	Range
Temperature (℃)	17.64	3.91	10.2 – 33.8
Humidity (%)	76.58	11.47	38.2 – 97.6
Ammonia (ppm)	5.9	4.45	0.1 – 31.7

Result-[1] Classification – Performance Metrics

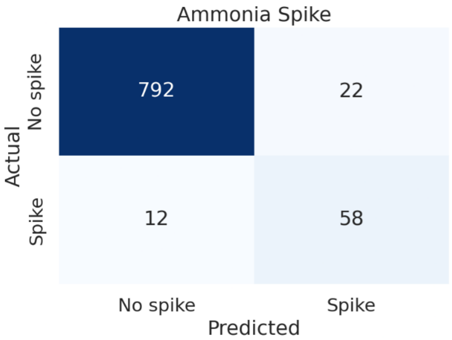
Metric	NH_3 spike	Temp spike
Accuracy	0.962	0.958
Precision	0.725	0.998
Recall	0.829	0.919
F1	0.773	0.957
AUC	0.982	0.996

Result-[2] Confusion Matrix – Temperature Spike



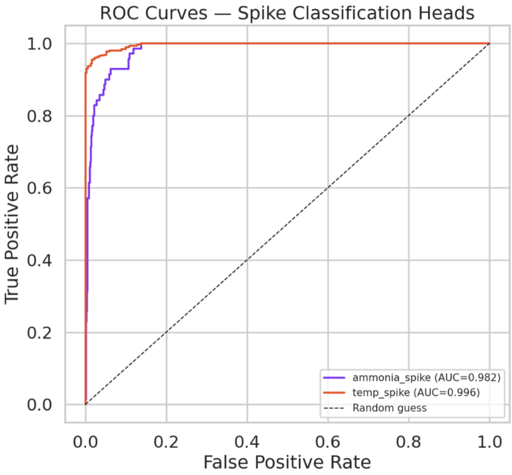
- TN: 440 – actual no spike, predicted no spike.
- TP: 407 – actual spike, predicted spike.
- FN: 36 – actual spike, missed.
- FP: 1 – false alarm.

Result-[3] Confusion Matrix – Ammonia Spike

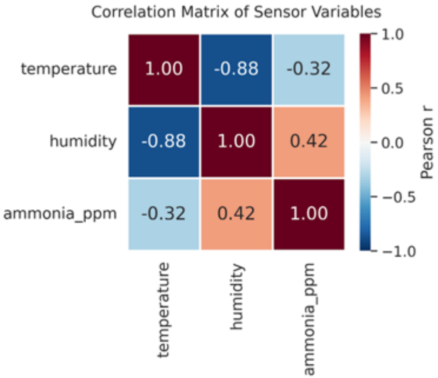


- TN: 792 – actual no spike, predicted no spike.
- TP: 58 – actual spike, predicted spike.
- FN: 12 – actual spike, missed.
- FP: 22 – false alarm.

Result-[4] Classification – ROC Curve

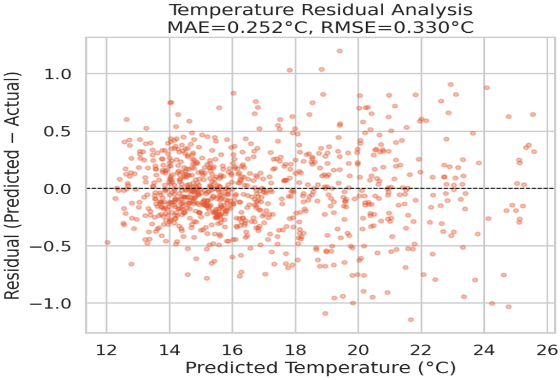


Result-[5] Correlation Analysis



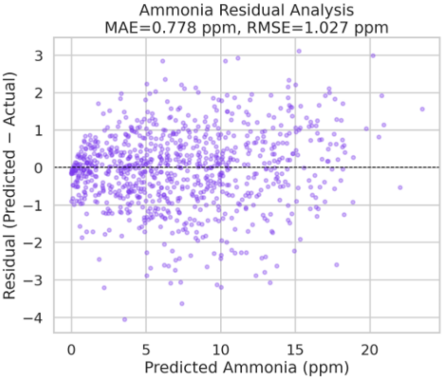
- Relationship between temperature and humidity is a strong negative correlation of -0.88.
- Ammonia and humidity has moderate correlation of 0.42.
- Higher humidity keeps litter, wet produces more ammonia.
- Ammonia and temperature has weaker negative correlation of -0.25.
- Negative relationship may be due to ventilation.

Result-[6] Residual Analysis – Temperature



- Positive residual shows actual value was higher than predicted.
- Negative shows actual value was lower than predicted.
- Temperature prediction is stable across the whole range.
- Residuals is centered on zero, no funnel shape, no curve.
- Error doesn't grow as predicted temperature climbs.

Result-[7] Residual Analysis – Ammonia



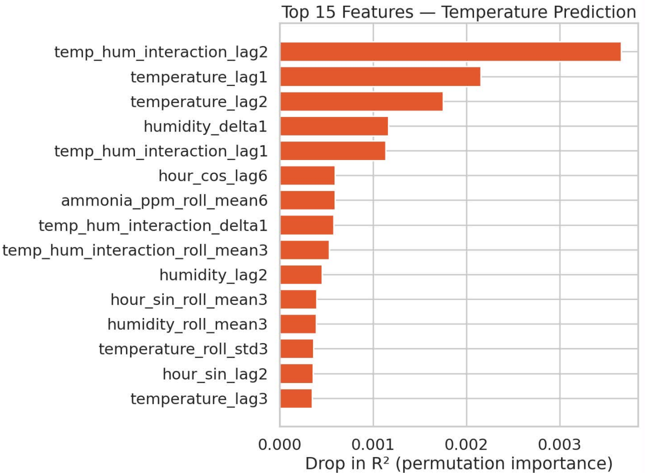
- Ammonia prediction is reliable at low concentrations but loses precision at higher
- Residuals fan out as predicted ammonia increases.
- Error grows with predicted ammonia, especially past 10 ppm, where some misses hit 3 ppm.
- Residuals stay centered on zero across the range.
- The model isn't consistently but less precise.

Result-[8] Regression Performance Metrics

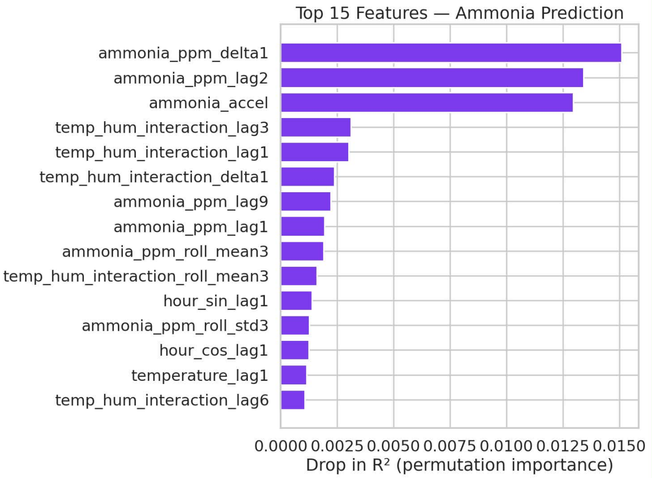
Metric	Temp	NH_3
MAE	0.252	0.778
MSE	0.109	1.054
RMSE	0.330	1.027
R^2	0.988	0.955

- On average, temperature predictions are off by 0.25℃.
- Model explains 98% of temperature variation.
- Model explains 95% of ammonia variation.

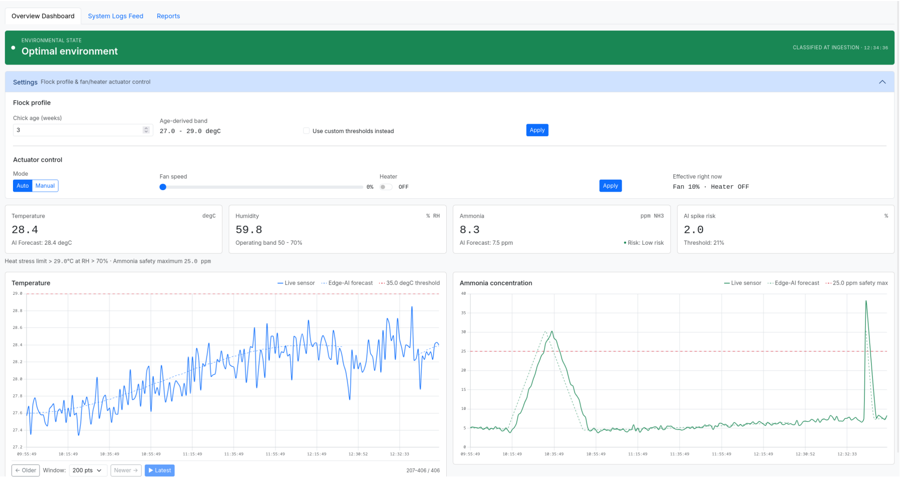
Result-[9] Feature Importance Analysis Temperature



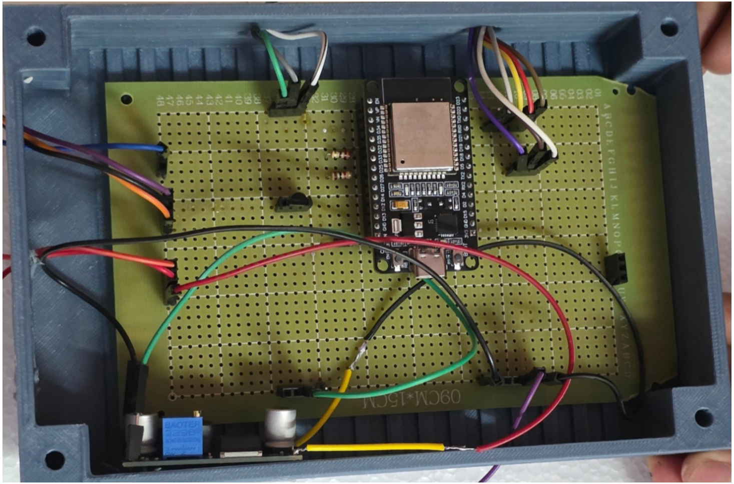
Result-[10] Feature Importance Analysis Ammonia



Result-[11] Monitoring Dashboard



Result-[12] Prototype



Result-[13] ESP-NOW Communication – Slave to Master

```
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
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[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
[TX] T=28.50C RH=72.00% NH3=0.00ppm -> master
[RX] Master classification: OPTIMAL_ENVIRONMENT  fan_pwm=0  heater_pwm=0
```

Result-[14] ESP-NOW Communication – Master Classification

```
[RX] seq=418 T=28.5C RH=72.0% NH3=0.0ppm
[MODEL] warming up -- not enough bucketed history yet (~70 min after boot).
[TX] Classification: OPTIMAL_ENVIRONMENT -- sending ActuatorCommand to sensor.
[ACTUATE] fan_pwm=0 heater_pwm=0 (state=OPTIMAL_ENVIRONMENT)
[RX] seq=419 T=28.5C RH=72.0% NH3=0.0ppm
[MODEL] warming up -- not enough bucketed history yet (~70 min after boot).
[TX] Classification: OPTIMAL_ENVIRONMENT -- sending ActuatorCommand to sensor.
[ACTUATE] fan_pwm=0 heater_pwm=0 (state=OPTIMAL_ENVIRONMENT)
[RX] seq=420 T=28.5C RH=72.0% NH3=0.0ppm
[MODEL] warming up -- not enough bucketed history yet (~70 min after boot).
[TX] Classification: OPTIMAL_ENVIRONMENT -- sending ActuatorCommand to sensor.
[ACTUATE] fan_pwm=0 heater_pwm=0 (state=OPTIMAL_ENVIRONMENT)
[RX] seq=421 T=28.5C RH=72.0% NH3=0.0ppm
[MODEL] warming up -- not enough bucketed history yet (~70 min after boot).
[TX] Classification: OPTIMAL_ENVIRONMENT -- sending ActuatorCommand to sensor.
[ACTUATE] fan_pwm=0 heater_pwm=0 (state=OPTIMAL_ENVIRONMENT)
[RX] seq=422 T=28.5C RH=72.0% NH3=0.0ppm
[MODEL] warming up -- not enough bucketed history yet (~70 min after boot).
[TX] Classification: OPTIMAL_ENVIRONMENT -- sending ActuatorCommand to sensor.
[ACTUATE] fan_pwm=0 heater_pwm=0 (state=OPTIMAL_ENVIRONMENT)
[RX] seq=423 T=28.5C RH=72.0% NH3=0.0ppm
[MODEL] warming up -- not enough bucketed history yet (~70 min after boot).
[TX] Classification: OPTIMAL_ENVIRONMENT -- sending ActuatorCommand to sensor.
[ACTUATE] fan_pwm=0 heater_pwm=0 (state=OPTIMAL_ENVIRONMENT)
[RX] seq=424 T=28.5C RH=72.0% NH3=0.0ppm
[MODEL] warming up -- not enough bucketed history yet (~70 min after boot).
[TX] Classification: OPTIMAL_ENVIRONMENT -- sending ActuatorCommand to sensor.
[ACTUATE] fan_pwm=0 heater_pwm=0 (state=OPTIMAL_ENVIRONMENT)
[RX] seq=425 T=28.5C RH=72.0% NH3=0.0ppm
[MODEL] warming up -- not enough bucketed history yet (~70 min after boot).
[TX] Classification: OPTIMAL_ENVIRONMENT -- sending ActuatorCommand to sensor.
[ACTUATE] fan_pwm=0 heater_pwm=0 (state=OPTIMAL_ENVIRONMENT)
[RX] seq=426 T=28.5C RH=72.0% NH3=0.0ppm
[MODEL] warming up -- not enough bucketed history yet (~70 min after boot).
[TX] Classification: OPTIMAL_ENVIRONMENT -- sending ActuatorCommand to sensor.
[ACTUATE] fan_pwm=0 heater_pwm=0 (state=OPTIMAL_ENVIRONMENT)
[RX] seq=427 T=28.5C RH=72.0% NH3=0.0ppm
[MODEL] warming up -- not enough bucketed history yet (~70 min after boot).
[TX] Classification: OPTIMAL_ENVIRONMENT -- sending ActuatorCommand to sensor.
[ACTUATE] fan_pwm=0 heater_pwm=0 (state=OPTIMAL_ENVIRONMENT)
[RX] seq=428 T=28.5C RH=72.0% NH3=0.0ppm
[MODEL] warming up -- not enough bucketed history yet (~70 min after boot).
[TX] Classification: OPTIMAL_ENVIRONMENT -- sending ActuatorCommand to sensor.
[ACTUATE] fan_pwm=0 heater_pwm=0 (state=OPTIMAL_ENVIRONMENT)
[RX] seq=429 T=28.5C RH=72.0% NH3=0.0ppm
[MODEL] warming up -- not enough bucketed history yet (~70 min after boot).
[TX] Classification: OPTIMAL_ENVIRONMENT -- sending ActuatorCommand to sensor.
[ACTUATE] fan_pwm=0 heater_pwm=0 (state=OPTIMAL_ENVIRONMENT)
```

Discussion

1. **Strong Model Performance**
    - Temperature: R^2 = 0.988; Ammonia: R^2 = 0.955.
    - Predictions closely follow the actual environmental patterns.
2. **Low Prediction Error**
    - Temperature: MAE = 0.252, MSE = 0.109.
    - Ammonia: MAE = 0.778, MSE = 1.054.
3. **Early Hazard Prediction**
    - ML predicts changes in temperature and ammonia before conditions become critical.
    - Supports early decision-making and proactive control.
4. **Real-Time Automated Monitoring**
    - Continuous monitoring of temperature, humidity, and ammonia.
    - Combines sensing, prediction, and automated control for faster response.

Future Enhancements

- **Additional Environmental Parameters:** CO_2 concentration, fan speed, airflow, and feed/water consumption for a more complete picture of ventilation and flock health.
- **Diverse and Larger Dataset:** data from more poultry houses, seasons, flock sizes, and ventilation types to improve generalization.
- **Disease and Behavior Monitoring:** microphone and camera-based detection of abnormal vocalization, movement, and feeding behavior.
- **Enhanced Sensor Technology:** more accurate, energy-efficient sensors to replace the MQ-137/DHT22 for better data quality.

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